Week 7 Lab: Post-Training, Forgetting, Model Editing and Merging

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Exercise 1

A foundation (base) model can be post-trained in three styles: *pre-train style*, *SFT style* and *RL style*.

(a) For each style, describe what one training example looks like, using the BeNeFri example from the slides: "The BeNeFri programme consists of the universities of Bern, Neuchâtel and Fribourg".
(b) A company has 10,000 internal documents, but no questions or answers about them. Which style can it use directly? What would it need to collect for the other two?
(c) Which style is usually applied to a *base* model, and which to an already aligned *chat/instruct* model? Why does post-training a chat model need extra care?

Exercise 2

(a) LLaMA-2-Chat was post-trained on Chinese data. Afterwards it answered a request for help to obtain bank passwords with instructions for an attack, while the original model refused. What was gained, what was lost, and why?
(b) A text LLM is fine-tuned with an adapter and a speech encoder on 23 speech tasks. After 1 epoch it still answers a "reply in JSON format" instruction correctly; after 3 epochs its answers are more accurate but it ignores the JSON instruction. No JSON data were in the fine-tuning set. Explain.

Exercise 3

LoRA freezes a pretrained weight $W \in \mathbb{R}^{d\times d}$ and learns $W + BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times d}$, initialised with $B = 0$ and random A.

(a) For $d = 4096$ and $r = 8$, how many parameters are trained per matrix, compared with full fine-tuning? What fraction is this?
(b) Why is B initialised to zero?
(c) The slides summarise LoRA as "learns less and forgets less". Explain both halves with the picture of the restricted region of reachable models. Why can such a region be "too narrow to achieve reliability and generality" for some updates?

Exercise 4

Model editing changes one specific piece of knowledge (e.g. "The Space Needle is in Seattle" → "... in Zurich").

(a) The slides distinguish editing *without* parameter modification from editing *with* parameter modification, and the latter into human-designed methods and "AI editing". Give an example of each.
(b) Why not simply fine-tune on the single sentence "The Space Needle is in Zurich"? Use the criteria of Exercise 2.

Exercise 5

Model merging combines fine-tuned models without training data or training. In the slides' example, one neuron of a foundation model has weights $\theta = (1, 2, -1)$. Fine-tuning for task A gives $\theta_A = (1, 2, -2)$ and for task B gives $\theta_B = (3, 2, -1)$.

(a) Compute the task vectors $\tau_A = \theta_A - \theta$, $\tau_B = \theta_B - \theta$ and the merged weights $\theta_A + \tau_B$. Why is the merge harmless here?

(b) Now suppose $\theta_B = (1, 2, 1)$ instead. Compute the merged weights again. What goes wrong? (The slides call these "inconvenient" parameters.)
(c) Why must θ_A and θ_B come from the same foundation model?

Exercise 6

Task vectors support three kinds of arithmetic: *addition*, *subtraction* (negation) and *analogy (proportion)*.

(a) LLaMA-2-Chat is $\theta_A = \theta + \tau_A$ (alignment) and a model trained on Chinese data from LLaMA-2-base is $\theta_B = \theta + \tau_B$. Which merge gives a model that speaks Chinese *and* stays safe? Why does this avoid the forgetting of Exercise 3(a)?
(b) You fine-tuned a model on toxic text, giving τ_{tox}. How can you use it to make a model *less* toxic?

Exercise 7

(a) Give two reasons why model merging is "not always successful".
(b) One line of work modifies the post-training loss so that the resulting models merge more easily. Suggest one property of the fine-tuned models that would make merging work better, and how a loss could encourage it.
(c) Compare four ways to add a skill without forgetting: LoRA, experience replay (with self-generated data), model editing, and model merging. For each, state what extra data or computation it needs.